Applied Physics A (2024) 130:231
https://doi.org/10.1007/s00339-024-07369-y

Applied Physics A
Materials Science & Processing

Crystal structure, Hirshfeld topology analysis, enrichment ratio, energy framework and DFT computational studies of piezoelectric L-histidinium tetrafluoroborate single crystal

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Received: 15 September 2023 / Accepted: 5 February 2024 / Published online: 15 March 2024

Abstract

The nonlinear L-histidinium tetrafluoroborate (LHFB) single crystals have been grown by the slow evaporation method. The single-crystal XRD confirms the monoclinic lattice system with the $P2_1$ space group and explains molecular electron charge density distribution for grown crystals. The diffraction data are refined to a final Refinement factor (R) value of 0.04. The growth rate of LHFB crystal is determined by a morphological study. The 3D Hirshfeld surface gives information about the intermolecular interaction present and the 2D fingerprint plot shows the nature of intermolecular close contact experienced by the molecules in a crystal. Energy frameworks have been computed by the Hirshfeld topology. The value of the enrichment ratio has been deduced to help understand the crystal packing interaction. 3D energy framework is a new way for understanding the topology of all types of interaction of the molecule in the crystal. The lower value of the dielectric loss at higher frequencies has increased the optical quality of the crystal. The piezoelectric property was reported for the first time in LHFB single crystal. The piezoelectric charge coefficient (d_{33}) is found to be 4 pC/N. The promising piezoelectric and dielectric studies of grown crystal make it suitable for various optoelectronics, energy harvesting, piezoelectric transducers and patch antennae for wireless communication applications. Mechanical properties reveal that the LHFB crystals belong to the softer nature. The voids present in the grown crystal were estimated at different isosurface values, which give information about the mechanical strength and porosity of the compound. The computing of geometric parameters, optimization energies, frontier molecular orbital energies and molecular surface electrostatic potential were performed using the Gaussian 09 program package in the DFT/B3LYP/6-311++G(d,p) basis set.

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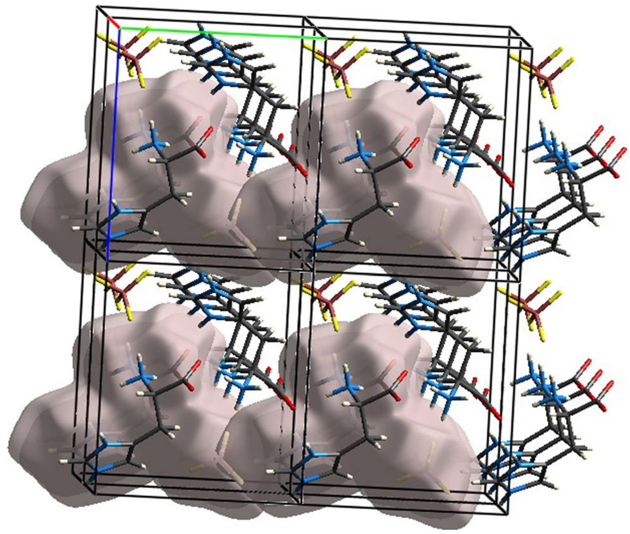
Graphical abstract



Keywords Crystal structure · Dielectric property · Piezoelectric materials · DFT computational studies

1 Introduction

The discovery of piezoelectricity in single crystals based on amino acids opens new avenues in the realms of electronics and biomedical fields. Understanding the piezoelectric characteristics of single crystals of L-amino acids require intense research. These crystals' versatility has technological significance as because of their valuable nonlinear optical properties [1, 2]. Amino acids are ideally suited for energy harvesting applications due to their low dielectric value. The tetrafluoroborate (BF4 −) ion from fluoroboric acid and the amino group (NH3+) from L-histidine are combined in LHFB material, highlighting its potential for effective energy consumption [3]. N atoms of the amino group undergo protonation, facilitated by the transfer of protons from tetrafluoroboric acid and carboxylic acid. This protonation occurs concurrently with the presence of carboxylate and tetrafluoroborate anion [4]. The occurrence of nonlinearities in LHFB is attributed to delocalized electrons linked to the histidine imidazolium and carboxylate functionalities [5, 6]. The other existing L-histidinium complexes are salts of L-histidine acetate [7], L-histidine oxalate [8], L-histidine maleate [9], L-histidinium tetrafluorophthalate [10], L-histidine monohydrochloride [11], L-histidinium dihydrogen arsenate orthoarsenic acid [12] etc. N. Sinha et al. have presented findings on the nonlinear and optical properties of LHFB crystals [13]. Nevertheless, electron density, enrichment ratio, energy framework, and piezoelectricity remain unreported to date in the literature.

An advanced and high-speed computation single-crystal X-ray diffraction technique provides insights into both molecular and crystal structures of LHFB single crystals. This method serves as a distinctive tool for determining charge density distribution, spatial distribution [14] and residual density [15] in molecular crystal. The electron density distribution is a crucial observable for interpreting matter properties. The structure factor Fobs is determined by the intensity, and the spatial distribution of diffracted X-rays yields valuable information. Residual density, indicative of refinement quality, reflects the disparity between modeled and observed electron density. The growth rates and morphological studies of LHFB crystals were assessed using the Bravais–Friedel–Donnay–Harker (BFDH) law [16]. Morphological simulations were based on crystal lattice geometry. A crystal's shape aims to minimize total surface energy for a given volume, considering the surface energies of the grown crystal. According to this law, a high interplanar distance (dhkl) corresponds to a high morphological importance (MI) for the hkl face. The crystal structure provides insights into intermolecular contacts and crystal packing for LHFB crystals, contributing crucial information.

In crystal engineering, the interaction between molecules and ions within crystals holds significant importance. Analyzing the Hirshfeld surface provides valuable insights into the intermolecular interactions present in grown crystals [17]. Utilizing the energy framework as a robust tool allows for the calculation of interaction energy between pairs of molecules, facilitating the visualization of the crystal structures' supramolecular architecture and generating a

3-dimensional topology of the prevailing crystal packing. The Hirshfeld surface analysis data enables the determination of the 'Enrichment Ratio,' indicating the propensity of chemical species to pair through crystal packing interactions. Furthermore, it is noteworthy that piezoelectricity is correlated with proton ordering in the amino acid structure [18, 19], emphasizing its connection to the microscopic solid structure. A computational study using density functional theory (DFT) has also been laid out in this work to understand the structural properties, molecular and electronic interactions of LHFB.

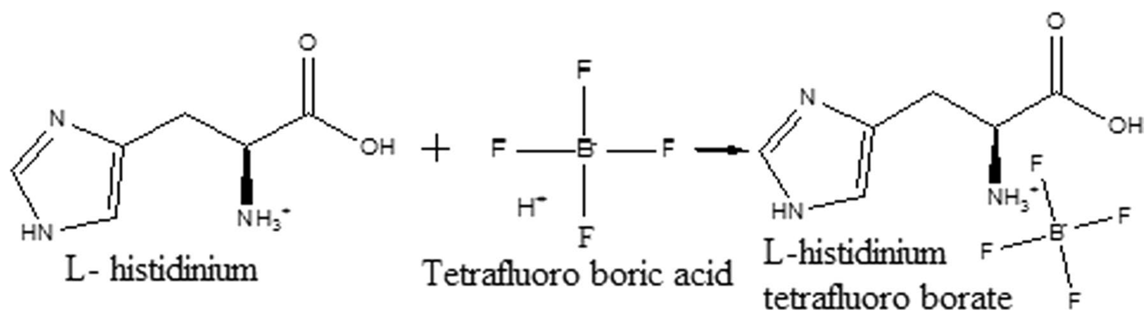
In this study, density functional theory (DFT) was employed to comprehensively investigate the structural properties, molecular interactions, and electronic characteristics of LHFB. Utilizing a theoretical model based on the B3LYP/6-311++G(d,p) basis set, our analysis covers optimized geometry, molecular electrostatic potential, frontier molecular orbital, and HOMO–LUMO analysis [20]. The successful preparation of LHFB single crystals was achieved through a meticulous slow evaporation process. This work includes a thorough examination of crystal structure and morphology, with molecular crystal contacts analyzed using Hirshfeld topology. The enrichment ratio provides valuable insights into crystal packing interactions. In addition, the energy framework is presented in terms of electrostatic, dispersion, and total energy.

Furthermore, this study reports on the crystal growth methodology and presents a detailed exploration of the contacts within the molecular crystal structure using Hirshfeld topology analysis. The enrichment ratio sheds light on the formation of crystal packing interactions. The energy framework is elucidated in terms of electrostatic, dispersion, and total energy. Moreover, the work delves into the FTIR, dielectric, piezoelectric, and mechanical properties of LHFB crystals, complemented by DFT studies.

2 Experimental

2.1 Crystal growth and morphology

L-Histidinium tetrafluoroborate has been synthesized from L-histidinium and tetrafluoroboric acid mixed in 1:1 ratio in distilled water as solvent [13]. The schematic representation of synthesis reaction is



The saturated solution of LHFB material was covered with aluminum foil and kept in a constant temperature bath for controlled slow solution evaporation. Transparent, colorless crystals were obtained of size 5 mm × 3 mm × 2 mm were harvested after 25–30 days [21]. The crystal is a polyhedron with developed faces (010), (001), (0–11), (1–21). The crystal shape is governed by the relative growth rates of crystal faces. The pictorial representation is represented in Scheme 1. A photograph, morphology, and plane of LHFB crystals under the experimental conditions have been shown in Fig. 1.

The simulation of crystal morphology is computed by Bravais–Friedel–Donnay–Harker (BFDH) law. It uses as a quick tool to approximate the habit of a crystal. The rate of growth of crystal plane BFDH law [22] is given as

$$\frac{\mathrm{d}l_{hkl}}{\mathrm{d}t} = \frac{R_{h_1k_1l_1}\sin\gamma + R_{h_2k_2l_2}\sin\alpha - R_{hkl}\sin(\alpha+\gamma)}{\sin\alpha\sin\gamma}, \quad (1)$$

where R_{hkl}, $R_{h_1k_1l_1}$ and $R_{h_2k_2l_2}$ are the growth rates of respected faces and α and γ are the interfacial angles. The calculated morphological importance (MI) of LHFB crystal is directly reflected to its corresponding area of crystal faces [12, 23]. The crystal packing of LHFB crystal in a BFDH relative area of different faces [24] is shown in Fig. 1b. The morphological importance and growth rate for LHFB crystal are shown in Table 1.

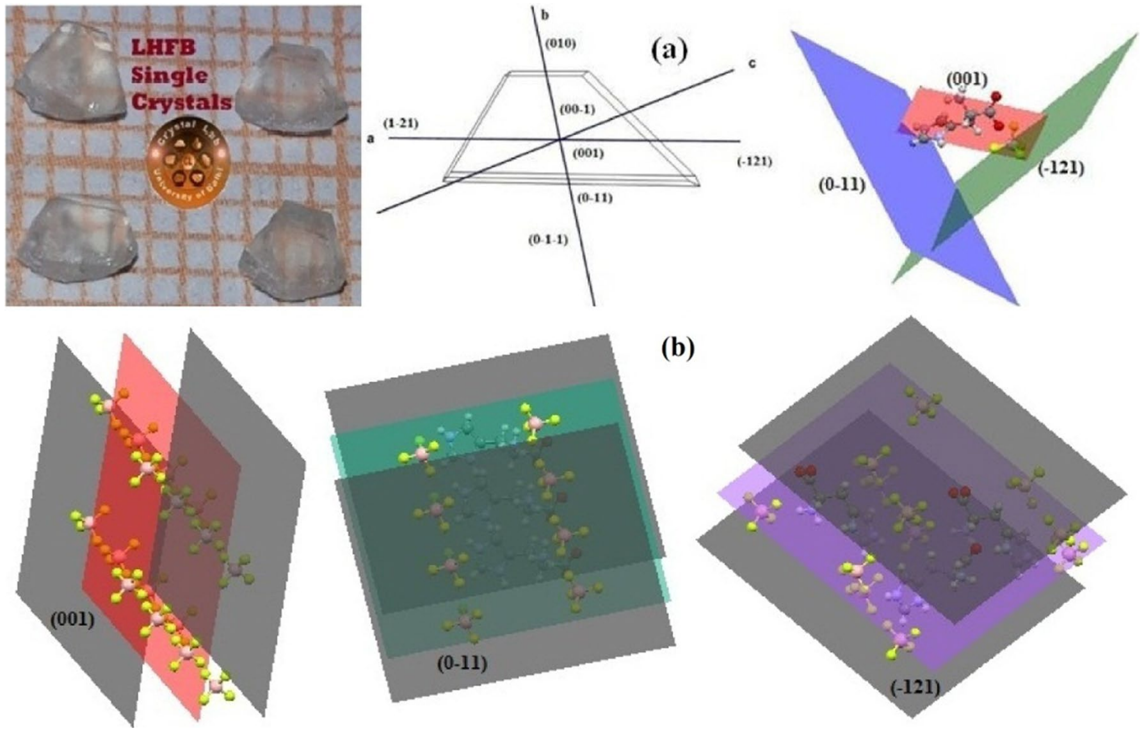


Fig. 1 **a** Photograph and crystal morphology of as-grown LHFB crystal. **b** Crystal packing and its relative growth for different crystal

Table 1 Morphological importance and growth rate for LHFB crystal faces using modified BFDH law

Faces (*hkl*)	d_{hkl} (Å)	$\frac{dl_{hkl}}{dt}$	Morphological importance by modified BFDH law
$(\bar{1}21)$	3.684	0.669	1.000
$\left(0\bar{1}1\right)$	6.785	0.539	1.241
(001)	10.206	0.421	1.589

3 Results and discussion

3.1 Crystal structure and X-ray diffraction

The crystal structure of LHFB crystal was determined by single-crystal X-ray diffraction. The crystallographic data were solved and refined by WinGx software using SHELXS 97 [25]. The H atom was refined with isotropic displacement parameters. The crystal structure consists of a five-membered lactone [-C4-C5-C6-N2-N3)] with a –CH_2 group and (CO_2) (NH_3) attached to the C2 unit. Boron is found to be tetrahedrally coordinated by four fluorine atoms in the LHFB structure. The one –CH_2– groups (C3) and one tertiary CH (C2) are placed geometrically using standard methods. N1 is attached to a conjugated system. The refinement of the LHFB structure in the space group $P2_1$ was proceeded straightforwardly fashion to a value of $R = 0.0375$ and Goodness of Fit (S) = 1.097. The experimental and crystal data are given in Table 2.

The single-crystal structure of L-histidinium tetrafluoroborate using Platon with 50% probability has been shown in Fig. 2.

Powder X-ray diffraction (XRD) measurement on LHFB material was performed using Bruker D8 Advanced X-ray diffractometer. Figure 1 represents the indexed powder XRD pattern of LHFB crystal. Sharp Bragg's reflection confirms the crystalline nature of synthesized material. The titled crystal belongs to monoclinic lattice with non-centrosymmetric space group (P21).

3.2 Residual electron density

The residual electron density is method which help to collecting the information about the crystallographic analysis. This method is also used to separate noise from structural information on the residual density [26]. Application of charge density analysis is the evaluation of one electron property in molecular crystal. Residual electron density analysis can explain how to assess the three-dimensional distribution without loss of information. The residual electron density is defined as

Table 2 Crystal data and structure refinement for LHFB single crystal

Empirical formula	$C_6 H_{10} B F_4 N_3 O_2$
Formula weight	242.98
Temperature	303(2) K
Wavelength	0.7108 Å (Mo Kα radiation)
Crystal system, space group	monoclinic, $P2_1$
Unit cell dimensions	0.5 × 0.5 × 0.5 mm
Cell length a	5.007 Å (7)
Cell length b	9.092 Å (11)
Cell length c	10.214 Å (13)
Cell angle alpha	90.00°
Cell angle beta	93.329° (11)
Cell angle gamma	90.00°
Volume	464.22 (10) $Å^3$
Z, calculated density	2, 1.738 g/cm^3
Absorption coefficient	0.80 mm^{-1}
F (000)	248
Crystal size	0.5 × 0.5 × 0.5 mm^3
Theta range for data collection	2.99–29.16°
Limiting indices	$-6 <= h <= 3$, $-8 <= k <= 12$
	$-14 <= l <= 14$
Standard reflection	2177
Independent reflection	1563 (R (int) = 0 0.0375)
Data completeness	1.17/0.62
Absorption correction	multi-scan
Refinement method	Full-matrix least-squares on F^2
Goodness-of-fit on F^2	1.097
Final R indices [I > 2σ (I)]	R1 = 0.038, wR2 = 0.091
R indices (all data)	R1 = 0.038, wR2 = 0.092

$$\Delta\rho(r) = \rho\text{calc} - \rho\text{obs} = \frac{1}{V}\sum_{H}(F(H)\text{obs} - F(H)\text{cal})e^{-2\pi iH.r + j\varphi(\text{cal})}, \tag{2}$$

where structure factor amplitudes $F(H)_{\text{obs}}$, $F(H)_{\text{cal}}$ and phase φ(calc) are used in the Fourier summation, which is independent of the X-ray frequency [26]. F_{obs}–F_{cal} maps are obtained by SHELXL software which is represented in Fig. 3a. The residual density map in the plane of the molecule was obtained in the final cycle of refinement [26]. The minimum and maximum values for residual density were – 1.52 and 1.41 e $Å^3$, respectively. The residual electron density map, contour level at 0.1(residual density distribution intervals), zero contour (red dot lines), positive contour (blue line) and negative contour (green line) [24] are shown in Fig. 3b. The 3D surface and 3D isosurface residual electron density maps with zero contour levels are shown in Fig. 3 (c-d). The residual electron density distribution is calculated to draw a histogram which is compared with a Gaussian distribution [28]. Figure 3e shows the residual density histogram for a refinement of LHFB crystal data. In Fig. 3e, residual density values are close to zero at the higher frequencies. For better visualization, we have used the logarithm of this histogram. In the logarithm scale (Fig. 3f); residual density distribution plot shows the deviation from the parabolic shape and the presence of systematic errors [28].

3.3 Hirshfeld surface analysis

The Hirshfeld surface analysis provides us with a better understanding of the intermolecular interactions that occur in the grown crystals and are responsible for their design and crystal packing [29]. The Hirshfeld surface of the molecule is defined as the space occupied by a molecule in a crystal for partitioning the crystal electron density into molecular fragments [30].

The Hirshfeld surface is different from molecular surface in the sense that apart from the molecule itself, it also includes the proximity with the nearest neighbors [30]. Figure 4 shows the Hirshfeld surface with the 0.002 a. u. isosurface of electron density, deformation density, electrostatic potential and orbital plotted on the Hirshfeld surface.

Electrostatic potential surfaces [31] can be either displayed as isocontour surfaces or mapped onto the molecular electron density. The positive electrostatic potential shows hydrogen donor potential (blue region), while negative electrostatic potential (red region) represents hydrogen bond acceptors. The electrostatic potential and orbital on an isosurface of the electron density 0.002 a.u. are shown in Fig. 5a.

Hirshfeld surface was calculated using Crystal Explorer [32] software to identify the important intermolecular interaction within the crystal structure. The distance d_e and d_i on the Hirshfeld surface provides a three-dimensional picture of the intermolecular close contact in a crystal [33]. The distance parameter d_e and d_i are the distance from the surface to the nearest nucleus outside and inside the surface, respectively. These parameters help us in picturizing the intermolecular close contact in a crystal. The d_e surface-displayed here an intense red circle shows a hydrogen-bonding interactions with the carboxyl acceptor group. The 2D fingerprint plots of the Hirshfeld surface are sensitive to the immediate environment of the molecule. Each data point in the fingerprint plot was assigned a color based on the fraction of the surface area corresponding to d_e and d_i. The color-coded information represents the intermolecular interaction in the 2D fingerprint plot [35]. The intermolecular interactions contribution is increased from blue to red color by following the green color. The d_e, d_i surface mapping and d_{norm} distance for LHFB crystals are shown in Fig. 5b.

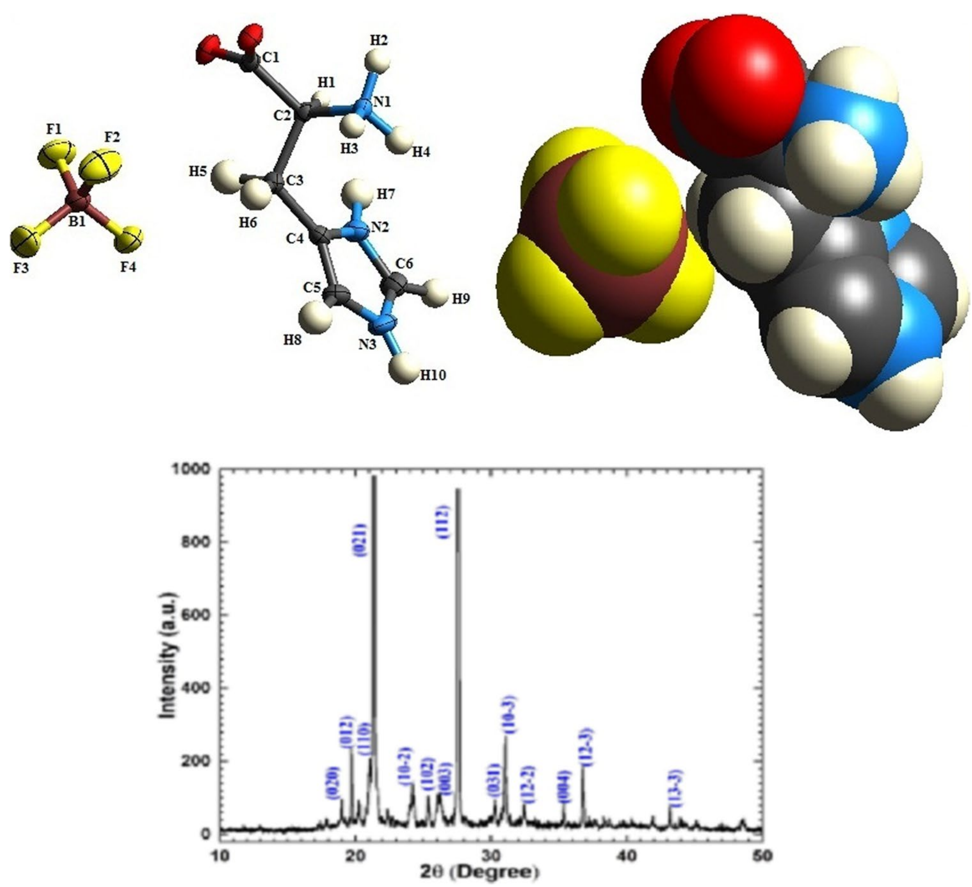


Fig. 2 Crystal structure for LHFB single crystal using Platon with 50% probability and crystal packing and Powder XRD plot of as-grown LHFB crystal

The inter-contacts, highlighted by surface mapping of d_{norm} on molecular Hirshfeld surfaces are shown in Fig. 6a. The red spot over the surface is indicated as the inter-contacts involved in the hydrogen bonds. d_{norm} surface helps to analyze the intermolecular interaction of LHFB crystal structure [34]. The hydrogen bonding in LHFB structure and distance of the nearest atom inside and outside surfaces have been shown in Fig. 6b.

Figure 7 shows that the breakdown of the fingerprint plot with different specific interaction (H…..F, H…..H etc.). The separation of contribution from different interaction types commonly overlaps in full fingerprint. The percentages of different intermolecular interactions present in the LHFB single crystal are visualized in Fig. 7.

By looking at the fragment patches on the Hirshfeld surface, we can conveniently identify the neighbor coordination environment of the molecule [Fig. 8a], if we know its coordination number. This is because the fragment patches on the Hirshfeld surface give measure of their closeness to the adjacent molecules [35]. The front and back views of the Hirshfeld surface mapped over fragment patches for L-histidinium tetrafluoroborate and hydrogen squarate mono-anion are separately shown in Fig. 8b.

3.4 3D-energy framework

The total interaction energy between LHFB compound is divided in to four parts: electrostatic, polarization, dispersion and exchange–repulsion. Mathematically, it can be written as [36]

$$E_{\text{total}} = k_{\text{ele}}E_{\text{ele}} + k_{\text{pol}}E_{\text{pol}} + k_{\text{dis}}E_{\text{dis}} + k_{\text{rep}}E_{\text{rep}}, \quad (3)$$

where classical electrostatic energy (E_{ele}) is obtained from the interaction between monomer charge distribution, polarization energy (E_{pol}) is approximated by adding all the atoms which have terms like $-\frac{1}{2}\,\alpha|F|^2$, dispersion energy ($E_{\text{dis}}$) is obtained by adding Grimme's D2 dispersion correction for all the pairs of atoms between the two molecules and exchange–repulsion energy is obtained from the antisymmetric product of the monomer spin orbital's. The scalar factors k_{ele}, $k_{\text{pol,}}$ k_{dis} and k_{rep} are determined by calibrating the values using the quantum mechanical results [36]. This energy was determined for molecules around the selected molecule at B3LYP/6-31G (d,p) level. The scale factors for CE-B3LYP benchmarked energy model are given in Table 3.

The energy frameworks were developed as red, green and blue colored cylinders, for E_{ele}, E_{dis} and E_{tot}, which represent the interaction energies for the title molecules (Fig. 9).

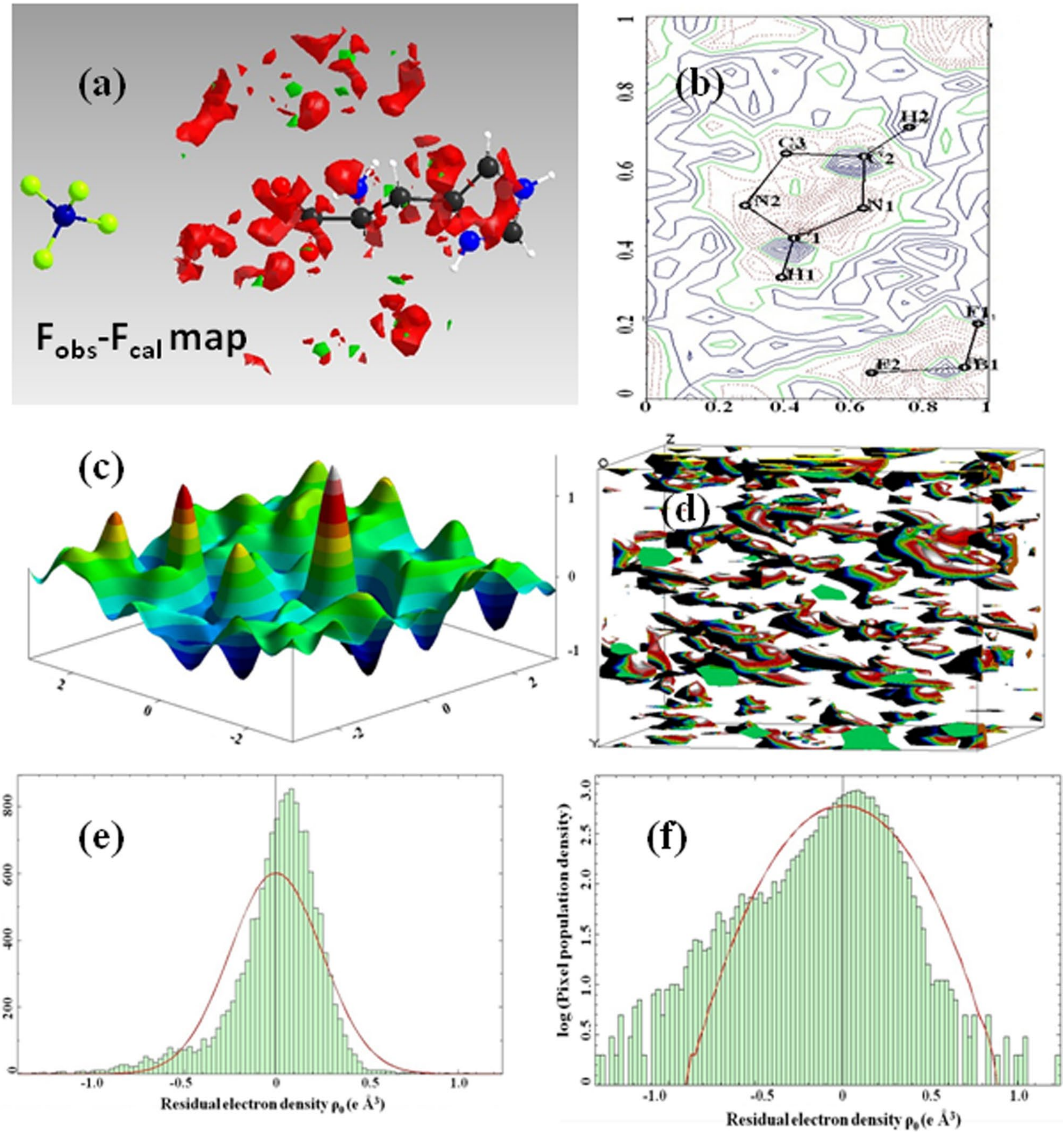


Fig. 3 **a** F_{obs}–F_{cal} map as mesh style isosurface. **b** Residual density map, the contour levels at 0.1 eÅ^{-3} intervals. **c**, **d** 3D surface and 3D isosurface residual electron density maps with zero contour level. **e** Histogram of the residual density. **f** The histogram of the residual density plotted on a logarithmic scale

These cylinders show the relative strength of molecular packing in different directions and its radius proportional to the magnitude of the interaction energy [35]. The value of interaction energies calculated by the energy model is shown in Table 4

3.5 Enrichment ratio

The ratio of actual contact C_{XX}/C_{XY} and random contacts R_{XX}/R_{XY} in the crystal is known as enrichment ratio [18].

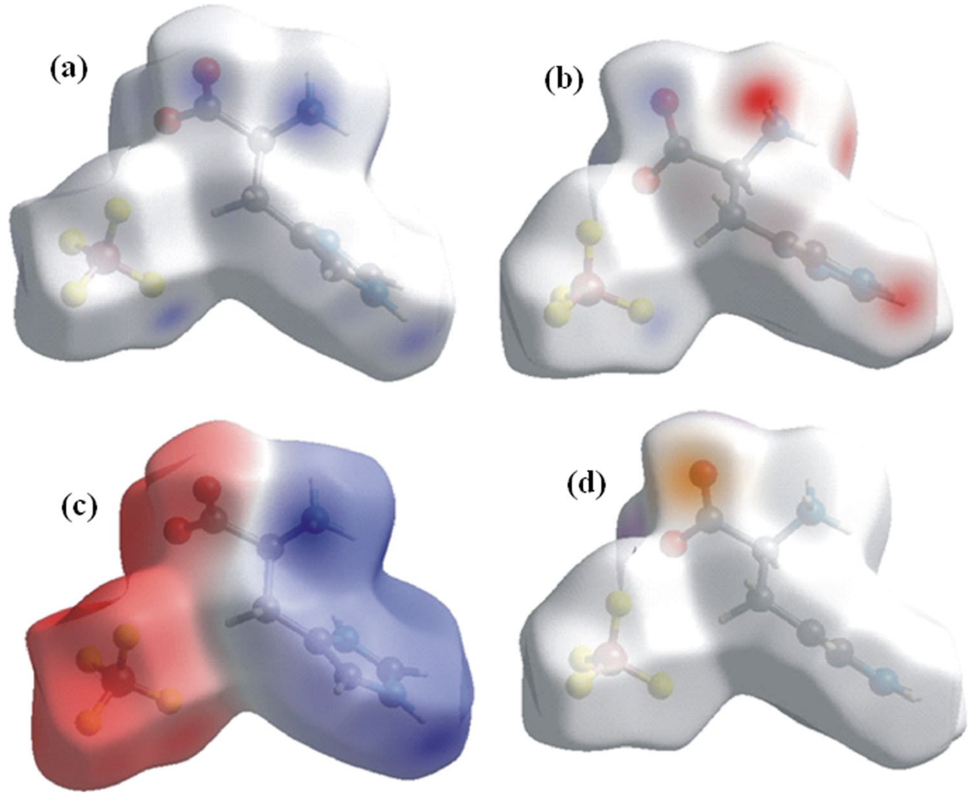


Fig. 4 Hirshfeld surface with the 0.002 a.u. isosurface of **a** electron density, **b** deformation density, **c** electrostatic potential and **d** orbital plotted

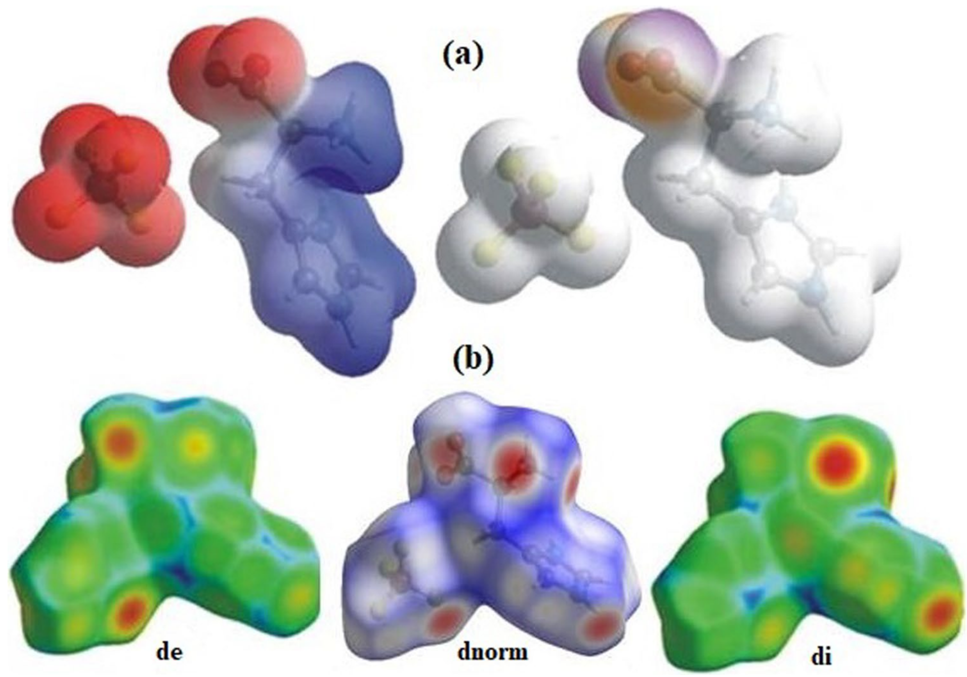


Fig. 5 **a** The experimental electrostatic potential and orbital on an isosurface of the electron density 0.002 a.u. The negative and positive d_{norm} values are represented by red and blue region in Hirshfeld surface. **b** d_e, d_i and d_{norm} of LHFB compound. Both d_e and d_i are responsible for d_{norm} distance

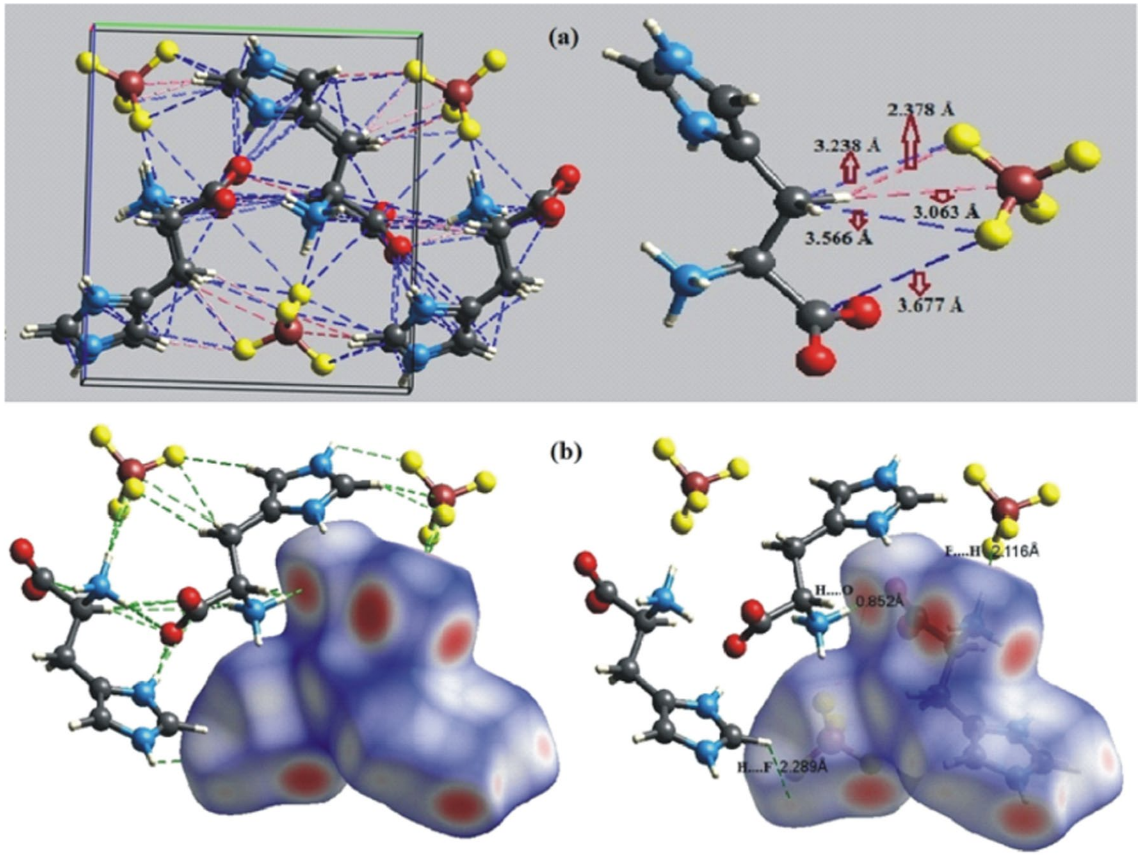


Fig. 6 **a**, **b** The hydrogen bonding and bond length of closest atom inside and nearest atom outside the surface of LHFB d_{norm} structure

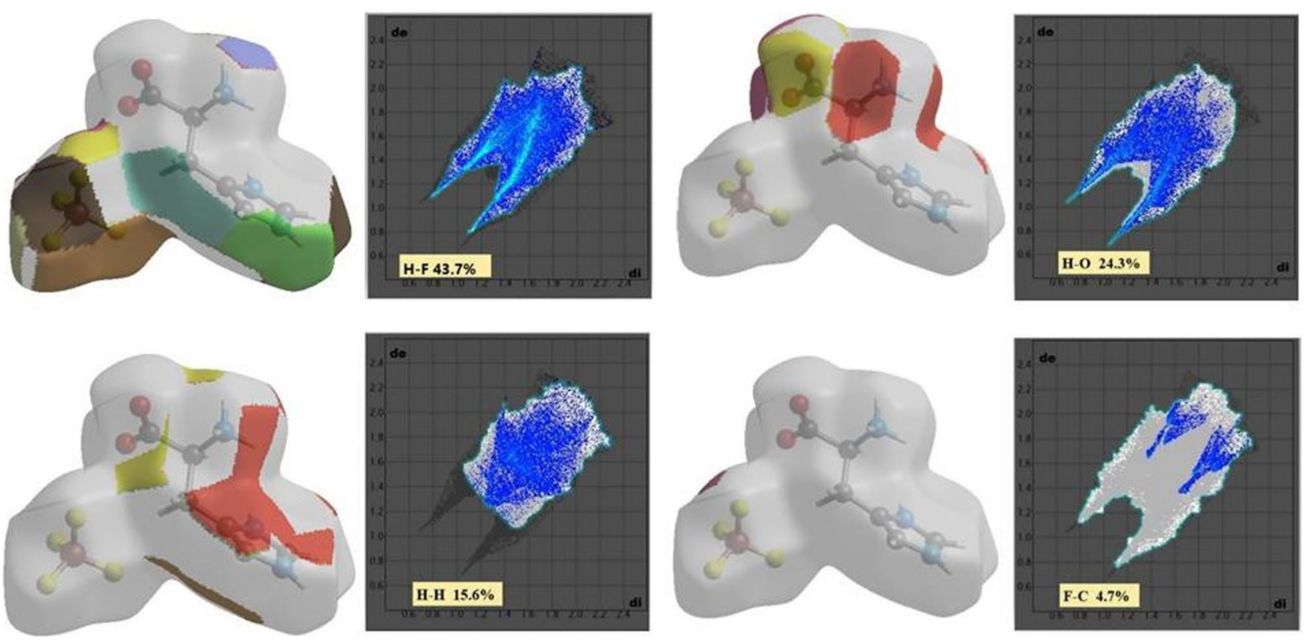


Fig. 7 The fingerprint plot of LHFB with specific interaction for different contacts

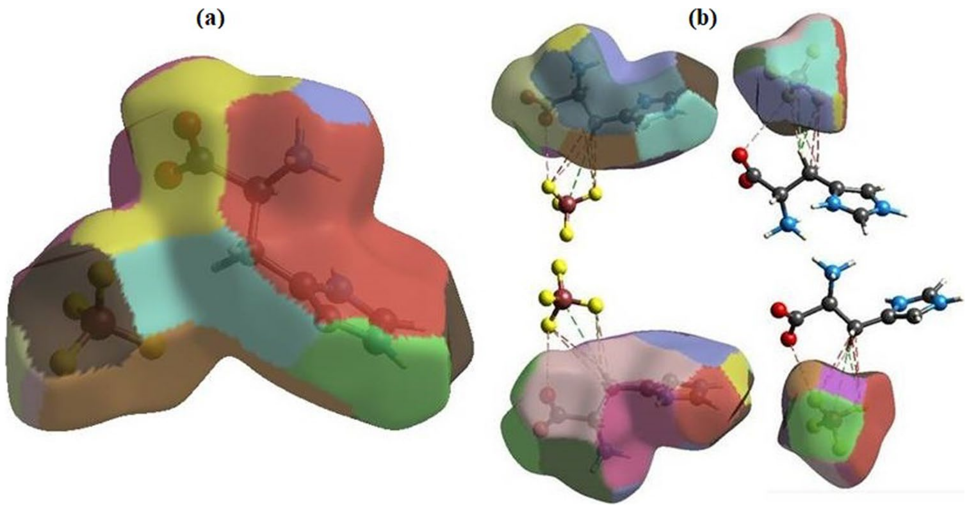


Fig. 8 **a** Fragment patch for LHFB molecules. **b** Front and back views of Hirshfeld surface over fragment patch for L-histidinium tetrafluoroborate and hydrogen squarate mono-anion

Table 3 Scale factor for benchmarked energy model

Energy model	k_{ele}	k_{pol}	k_{dis}	k_{rep}
CE-B3LYP.. B3LYP/6-31G (d,p) electron densities	1.057	0.740	0.871	0.618

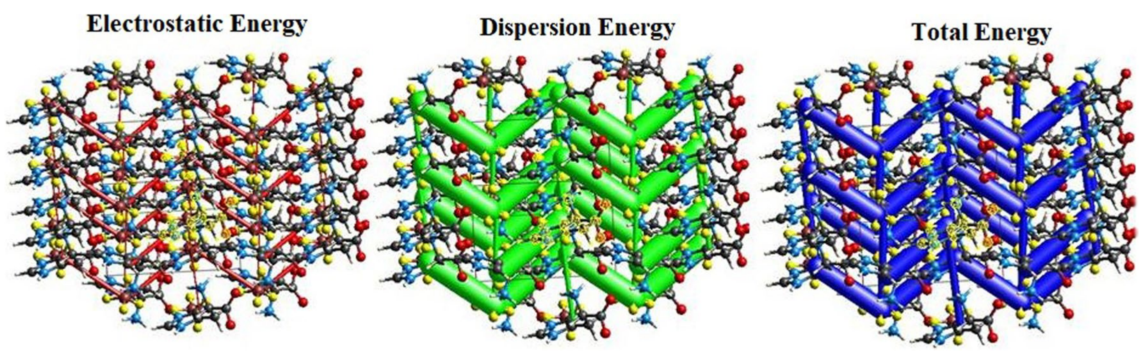


Fig. 9 Energy framework diagrams for electrostatic energy, dispersion energy and total energy for a molecule present in 3.8 Å cluster around selected reference molecule

Table 4 Interaction energies between a reference molecule and its neighbor

S. no	Number of equivalent neighbors (N)	Distance between molecular centroids (R)	E_{ele} $\frac{kJ}{mol}$	E_{pol} $\frac{kJ}{mol}$	E_{dis} $\frac{kJ}{mol}$	E_{rep} $\frac{kJ}{mol}$	E_{tot} $\frac{kJ}{mol}$
1	1	4.74 Å	– 2.0	– 1.8	– 11.0	8.0	– 8.1
2	1	6.72 Å	– 0.7	– 0.9	– 2.4	0.3	– 3.3
3	1	5.14 Å	– 6.6	– 8.0	– 15.7	27.5	– 9.6

The contacts percentage between molecules present in a crystal packing is computed from Crystal Explorer. The random contacts can be evaluated from the probability product of proportions S_X and S_Y [37]:

$$R_{XX} = S_X S_X \text{ and } R_{XY} = 2 S_X S_Y. \tag{4}$$

In addition, the proportion S_X or S_Y which is chemical type X or Y on the molecular surface is obtained by summation,

Table 5 The contacts percentage from fingerprint plot and random contacts percentage for LHFB single crystal

Atoms	H	C	N	O	F
Contacts (%)					
H	15.6	–	–	–	–
C	1.8	0.0	–	–	–
N	1.5	0.0	0.0	–	–
O	24.3	1.5	1.0	0.1	–
F	43.7	4.7	2.3	0.7	2.8
Surface (%)	51.3	4.0	2.4	13.9	28.9
Random contacts (r, %)					
H	26.3	–	–	–	–
C	4.1	0.1	–	–	–
N	2.5	0.2	0.1	–	–
O	14.3	1.1	0.7	1.9	–
F	29.6	2.3	1.4	8.0	8.4

Table 6 The enrichment ratio (E) values for LHFB single crystal

Atoms	H	C	N	O	F
H	0.59	–	–	–	–
C	0.44	–	–	–	–
N	0.61	–	–	–	–
O	1.7	1.4	1.49	–	–
F	1.47	2.03	1.65	0.1	0.3

$$S_X = C_{XX} + \frac{1}{2}\sum_{X \neq Y} C_{XY}. \tag{5}$$

The fraction of enrichment ratio (E_{XY}) is mathematically calculated by given the equation [37, 38]

$$E_{XY} = C_{XY}/R_{XY}, \tag{6}$$

where the value of $E > 1$ for a pair of elements with a higher propensity to form contacts and $E < 1$ for pairs which tend to avoid contacts with each other [38, 39]. The percentage of contacts and Random contacts for LHFB single crystal are shown in Table 5.

It is observed that the enrichment ratio values of polar contacts (H–O, H–F), C–O, C–F, N–O and N–F are favored in the crystal packing and other type contacts are contribute low or disfavored due to less than one or zero value of their enrichment ratio (Table 6).

3.6 Dielectric property

Dielectric studies have been performed using a computer-controlled E4980A LCR meter in the frequency range 20 Hz–2 MHz with a cooling/heating rate of 2 °C/ min. The measurement was carried out in the temperature range from 27 to 70 °C. Figure 10a, b shows the variations of dielectric constant and dielectric loss with the frequency. In Fig. 10a, b, the dielectric constant is found to fall exponentially as the frequency is increased. The value of ε' is high (88) at lower frequency because at lower frequencies, all the factors such as electronic, ionic, orientational and space charge polarization contribute to its value. However, at higher frequencies, the contribution of polarization from various parts of the crystal becomes weak, and the value of dielectric constant is low (34) [40]. The behavior of dielectric loss at lower frequency depends on the growth rate, defect, etc. Because of small

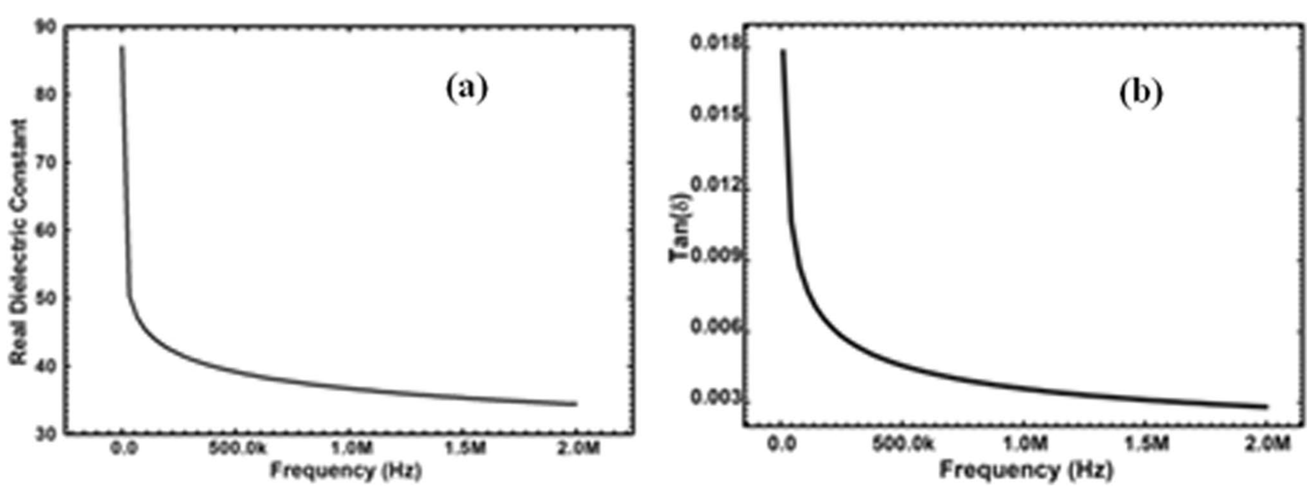


Fig. 10 **a**, **b** Variation of real parts of dielectric constant and dielectric loss versus frequency at room temperature. The high value of dielectric constant and dielectric loss are 88 and 0.017

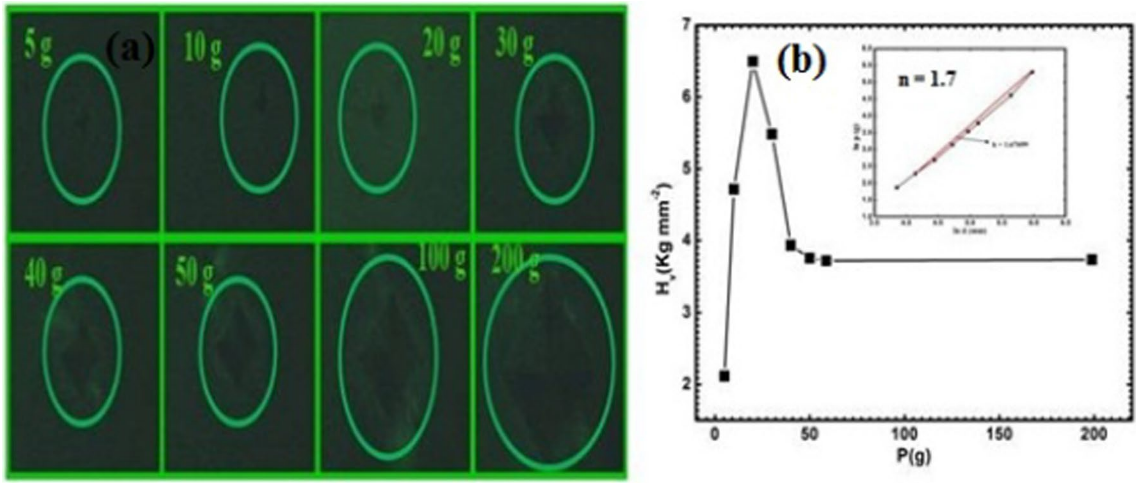


Fig. 11 **a** The Vickers indentation of grown crystal at various loads. **b** The variation of Vickers microhardness number (H_v) with load (P) for grown crystal insert figure shows work hardening coefficient calculated from P vs d plot

dielectric loss (0.003) at higher frequencies, LHFB crystals possess increased optical quality [41].

3.7 Piezoelectric property

Piezoelectricity in most of amino acid compounds have been related to the ordering of protons in their structure. It is a linear effect that is related to the microscopic structure of the solid. High piezoelectric coefficients are desirable in materials utilized as actuators [42].

Piezoelectric materials are a class of low-symmetry materials that can be polarized both by the application of an electric field and by mechanical stress. For piezoelectric charge coefficient (d) measurement, the crystal was coated with silver paste for electrical contact and a DC poling field 2.5 kV/cm was carried out on the crystal for 25 min. duration in the same direction in which d is measured. The piezoelectric coefficient value for the LHFB crystals was found to be 4pC/N. Due to piezoelectric properties, making them suitable for applications in piezoelectric transducer [42].

Fig. 12 Voids for LHFB crystal with four isovalues 0.001 a.u., 0.002 a.u., 0.003 a.u. and 0.005 a.u

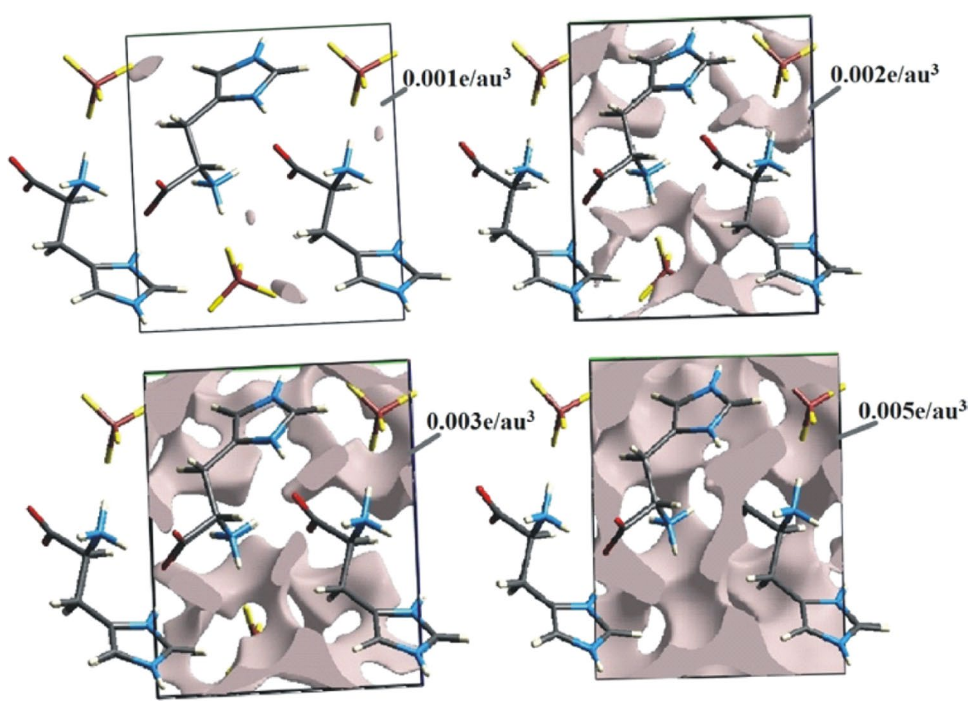


Table 7 Summarize the values of volume and surface area of the voids at different isovalues in the unit cell of the title compound

Isovalue (e/au^3)	Volume (Å^3)	Surface area (Å^2)
0.001	0.94	8.13
0.002	24.22	129.78
0.003	59.00	236.00
0.005	115.70	328.30

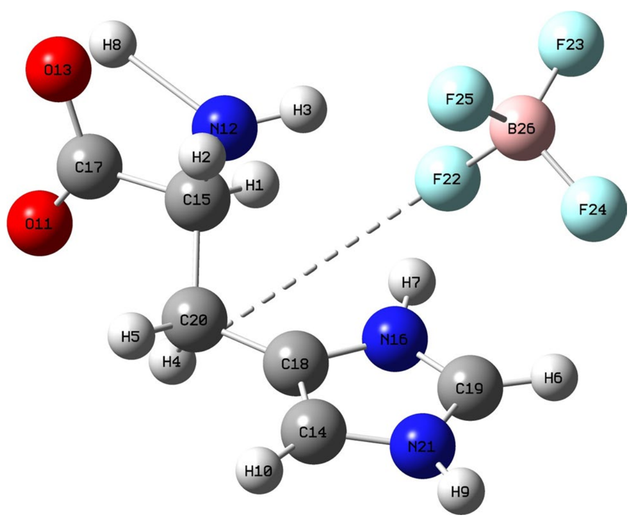


Fig. 13 Optimized geometry of LHFB

3.8 Microhardness study

The interactions and mobility of dislocations determine the hardness of a solid. It is a measure of resistance offers to local deformation. The material parameter called 'microhardness' measures the strength of the bond that forms the crystal structure. This property is related to the crystal structure of the material [43]. The indentation tests were carried out using the Vickers-pyramid method. Vickers hardness number (H_v) was calculated from the following relation:

$$H_v = \left[(2P \sin 136°)/2\right]/d^2 \quad (7)$$

$$H_v = 1.8544\left(P/d^2\right), \quad (8)$$

where P is the applied load and d is the average value of the indentation diagonals. The photographs of indentation marks on the LHFB crystal at various applied loads are shown in Fig. 11a. The load variation can be interpreted using Meyer's law [44]:

$$P = Kd^n, \quad (9)$$

where K is the material constant and n is the work hardening coefficient which is obtained from the plot of P vs d uses a least-square fit method (Fig. 11b). The value of 'n' should lie between 1 and 1.6 for hard materials and above 1.6 for softer ones. The value of n for LHFB was found to be 1.7 suggesting a softer nature of LHFB crystal.

The isosurface of pro-crystal electron density shows any void or cavity present in the crystal. It is an important component for discussing the structure of crystals. The void space in the crystal system is graphically presented by rolling a probe sphere of a variable radius over a Hirshfeld surface [45]. In the void surface, the sum of the pro-molecule density in the unit cell is less than its isovalue [34]. The void surface meets the boundary of the unit cell and capping faces are generated to create an enclosed volume. Figure 12 shows voids for LHFB crystal with four isovalues 0.001 a.u., 0.002 a.u., 0.003 a.u. and 0.005 a.u. The increasing isosurface value leads to a decrease in surface area and the volume of the voids (Table 7). An isovalue of 0.002 au is suitable for detecting voids and cavities in porous materials. The electron density contour at 0.002 a.u. covers more than 98% of the electronic charge [46].

3.9 Optimized geometry

We performed density functional theory (DFT) calculations with Gaussian 09 [46] using B3LYP exchange–correlation functional (Becke's three parameter exact exchange-functional (B3) combined with Lee–Yang–Parr correlation functional (LYP)) [47, 48]. The calculation of chemical/physical properties using DFT relies crucially on the accuracy of ground state parameters. In order to do this, first we carried out the geometry optimization of LHFB using the appropriate 6-311++G(d,p) basis set. The relaxation of structure was carried out by minimizing the total energy till the energy change of – 8.97e–7. The calculated value of this minimum energy is found to be – 26,507.86 eV. The dipole moment of the optimized structure is computed as 13.63 Debye with charge 0 and singlet spin. Figure 13 depicts the optimized geometry of LHFB and their structural parameters including bond lengths, bond angles and dihedral angles are listed in Table 8. We have used the optimized structure to calculate other properties such as HOMO–LUMO energy diagram, molecular electrostatic potential (MEP) map and frontier molecular orbitals etc.

The optimized parameters, viz. bond length, bond angle and dihedral angle are tabulated in Table 8. The calculated values of these parameters match reasonably well with experimental results. The intermolecular H4–F22 bond distance is found to be 4.20 A° along with bond angles 69.04° (C20-H4-F22) and 132.55° (H4-F22-B26). The bond

Table 8 Optimized bond length (A°), bond angle (°) and dihedral angle (°) of LHFB using B3LYP/6-311++G(d, p)

Bond length (A°)		Angle (°)		Dihedral angle (°)	
C15-H1	1.10	C20-H4-F22	69.04	F22-H4-C20-H5	175.23
N12-H2	1.01	H2-N12-H3	109.49	F22-H4-C20-C15	57.75
N12-H3	1.02	H2-N12-H8	109.47	F22-H4-C20-C18	– 67.93
C20-H4	1.09	H2-N12-C15	113.25	C20-H4-F22-B26	15.48
H4-F22	4.20	H3-N12-H8	126.61	H2-N12-C15-H1	– 152.46
C20-H5	1.09	H3-N12-C15	112.74	H2-N12-C15-C17	94.43
C19-H6	1.08	H8-N12-C15	82.86	H2-N12-C15-C20	– 29.92
N16-H7	1.05	H10-C14-C18	130.92	H3-N12-C15-H1	– 27.43
N12-H8	1.91	H10-C14-N21	122.22	H3-N12-C15-C17	– 140.54
N21-H9	1.01	C18-C14-N21	106.85	H3-N12-C15-C20	95.10
C14-H10	1.08	H1-C15-N12	106.87	H8-N12-C15-H1	99.37
C17-O11	1.20	H1-C15-C17	104.65	H8-N12-C15-C17	– 13.74
C15-N12	1.46	H1-C15-C20	109.13	H8-N12-C15-C20	– 138.09
C17-O13	1.33	N12-C15-C17	110.10	H10-C14-C18-N16	– 179.57
C14-C18	1.37	N12-C15-C20	116.90	H10-C14-C18-C20	0.29
C14-N21	1.39	C17-C15-C20	108.49	N21-C14-C18-N16	0.22
C15-C17	1.55	H7-N16-C18	126.61	N21-C14-C18-C20	– 179.92
C15-C20	1.55	H7-N16-C19	122.45	H10-C14-N21-H9	– 1.49
C18-N16	1.38	C18-N16-C19	110.32	H10-C14-N21-C19	– 179.59
C19-N16	1.33	O11-C17-O13	123.60	C18-C14-N21-H9	178.69
C18-C20	1.49	O11-C17-C15	123.02	C18-C14-N21-C19	0.59
C19-N21	1.34	O13-C17-C15	113.38	H1-C15-C17-O11	80.36
B26-F22	1.47	C14-C18-N16	105.94	H1-C15-C17-O13	– 98.42
B26-F23	1.36	C14-C18-C20	130.59	N12-C15-C17-O11	– 165.12
B26-F24	1.41	N16-C18-C20	123.46	N12-C15-C17-O13	16.11
B26-F25	1.42	H6-C19-N16	124.74	C20-C15-C17-O11	– 36.04
		H6-C19-N21	127.56	C20-C15-C17-O13	145.19

Table 8 (continued)

Bond length (A°)		Angle (°)		Dihedral angle (°)	
		N16-C19-N21	107.59	H1-C15-C20-H4	– 59.26
		H4-C20-H5	106.10	H1-C15-C20-H5	– 174.37
		H4-C20-C15	107.79	H1-C15-C20-C18	63.92
		H4-C20-C18	110.33	N12-C15-C20-H4	179.38
		H5-C20-C15	109.72	N12-C15-C20-H5	64.27
		H5-C20-C18	108.08	N12-C15-C20-C18	– 57.44
		C15-C20-C18	114.50	C17-C15-C20-H4	54.21
		H9-N21-C14	126.05	C17-C15-C20-H5	– 60.90
		H9-N21-C19	124.64	C17-C15-C20-C18	177.39
		C14-N21-C19	109.28	H7-N16-C18-C14	– 172.02
		H4-F22-B26	132.55	H7-N16-C18-C20	8.11
		F22-B26-F23	110.14	C19-N16-C18-C14	– 0.98
		F22-B26-F24	105.55	C19-N16-C18-C20	179.15
		F22-B26-F25	105.53	H7-N16-C19-H6	– 3.68
		F23-B26-F24	113.30	H7-N16-C19-N21	172.84
		F23-B26-F25	112.99	C18-N16-C19-H6	– 175.16
		F24-B26-F25	108.80	C18-N16-C19-N21	1.35
				C14-C18-C20-H4	– 102.32
				C14-C18-C20-H5	13.27
				C14-C18-C20-C15	135.88
				N16-C18-C20-H4	77.51
				N16-C18-C20-H5	– 166.89
				N16-C18-C20-C15	– 44.29
				H6-C19-N21-H9	– 2.94
				H6-C19-N21-C14	175.19
				N16-C19-N21-H9	– 179.33
				N16-C19-N21-C14	– 1.20
				H4-F22-B26-F23	– 154.52

Table 8 (continued)

Bond length (A°)	Angle (°)	Dihedral angle (°)	
		H4-F22-B26-F24	82.84
		H4-F22-B26-F25	– 32.27

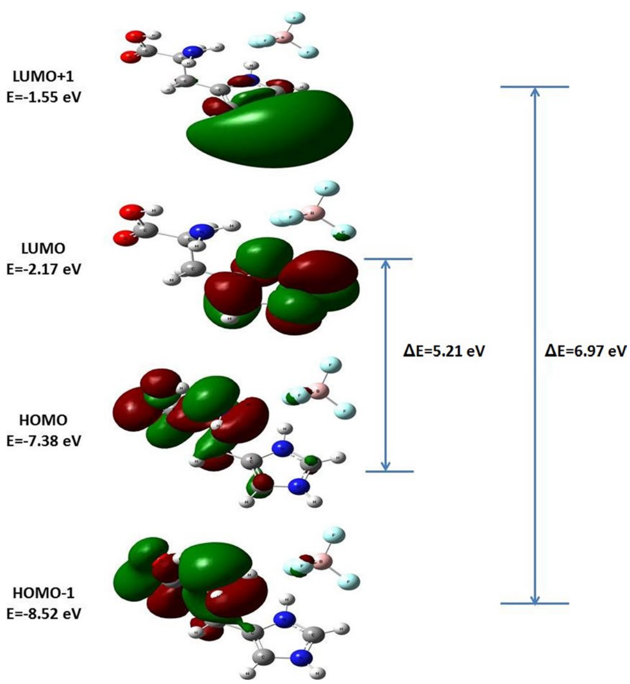


Fig. 14 Energy level diagram of LHFB using B3LYP/6-311++G(d,p) level of theory

distances of C20-H4 and B26-F22 in intra-molecular interactions are computed as 1.09 A° and 1.47 A° respectively. The bond lengths of B-F bonds are calculated as 1.47 A° (B26-F22), 1.36 A° (B26-F23), 1.41 A° (B26-F24) and 1.42 A° (B26-F25). The computed bond lengths of C–O bonds are found to be 1.20 A° (C17-O11) and 1.33 A° (C17-O13). The maximum bond lengths of C–N and N–H bonds are observed as 1.46 A° (C15-N12) and 1.91 A° (N12-H18) respectively. The intermolecular hydrogen bond interaction (H4–F22) between L-histidine (LH) and fluoroborate (FB) has been found in molecule which confirms the stability of LHFB.

3.10 Frontier molecular orbitals

Highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) are known as frontier molecular orbitals. FMO's provide vital information about the electronic and optical properties of the molecule.

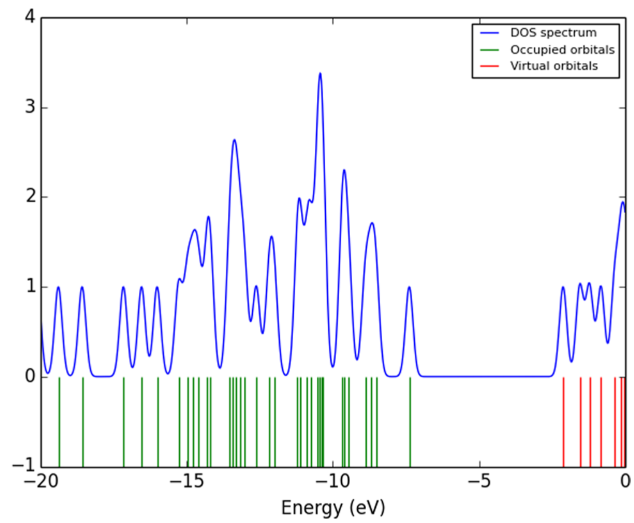


Fig. 15 Calculated density of states spectrum of LHFB

It can give insights about the potential of the molecule in luminescence, photochemical reaction, optoelectronics, biochemical, pharmaceutical related research. According to frontier molecular orbital theory, the chemical reactivity and stability of molecule can be determined by HOMO and LUMO energies [49, 50]. The HOMO–LUMO energy gap ($\Delta E = E_{LUMO}–E_{HOMO}$) is significant in understanding the molecular electrical transport properties i.e., charge excitation from HOMO to LUMO. Higher value of E_{HOMO} indicates that electrons will be more readily available from the molecule, whereas, lower value of E_{LUMO} shows the significant capability of molecule to accept electrons [49, 50]. The energies of HOMO and LUMO were computed using B3LYP/6-311++G(d, p) basis set. The calculated HOMO, HOMO-1, LUMO, and LUMO + 1 plot and their corresponding energies are displayed in Fig. 14. It can be seen in Fig. 14 that HOMO is localized on COO, NH_3 groups and slightly concentrated on CH_2 group of ring while LUMO is fully concentrated on imidazolidine ring due to delocalization of electrons. The same behavior can be found in HOMO-1 and LUMO + 1. HOMO–LUMO energy diagram shows that charge transfer happens within the molecule from COO and NH_3 groups to imidazolidine ring. The computed values of HOMO and LUMO energies are found to be – 7.38 eV and – 2.17 eV, respectively. The energy gap, $\Delta E = E_{LUMO}–E_{HOMO}$, between HOMO and LUMO was found to be 5.21 eV. A high value of energy gap reveals the high kinetic stability and low chemical reactivity of the molecule. The molecule has shown ability to participate in optical and luminescence activities due to its suitable HOMO–LUMO energy gap. This type of energy level diagram is probably better for such activities due to high value of energy gap and emission wavelength at about 238 nm. A combination

Table 9 The calculated values of quantum parameters of LHFB

Quantum parameters	Calculated values
Energy of HOMO, E_{HOMO} (eV)	– 7.38
Energy of LUMO, E_{LUMO} (eV)	– 2.17
Energy gap, ΔE (eV)	5.21
Electronegativity, χ (eV)	4.78
Chemical potential, μ (eV)	– 4.78
Chemical hardness, η (eV)	2.61
Chemical softness, S (eV)$^{-1}$	0.19
Electrophilicity index, ω (eV)	4.38

of high energy gap and charge transfer interaction with in molecule shows better NLO response of the molecule [51]

Gauss-sum program [52] is used to plot density of states (DOS) spectrum and molecular orbitals as shown in Fig. 15. Partial density of states (PDOS) spectrum plays a key role in illustrating the composition of molecular orbitals and their contributions to the chemical bonding. In Fig. 15, a positive value of DOS suggests a bonding interaction between orbitals/atoms while zero shows no interaction. Further, the calculated HOMO–LUMO energy gap is validated by this DOS spectrum.

Further, we have used E_{HOMO} and E_{LUMO} to calculate the global chemical reactivity parameters, viz. electronegativity (χ), chemical potential (μ), chemical hardness (η), chemical softness (S), global electrophilicity (ω), and global softness, for understanding the way the molecule interacts with other species. These are very useful to understand the chemical properties, stability and chemical strength of the molecule [53, 54]. The energies of HOMO and LUMO correspond to ionization energy (I) and electron affinity (A) of the molecule, respectively.

The electronegativity (χ) describes the tendency of an atom to attract electrons towards itself in a chemical bond and can be calculated as $\chi = -\left(E_{\mathrm{LUMO}} + E_{\mathrm{HOMO}}\right)/2$.

The global electrophilicity index (ω) measures the tendency of a chemical species to accept electrons which is defined as follows:

$$\omega = \frac{\mu^2}{2\eta}$$

where μ represents the chemical potential and η represents chemical hardness of the molecule. The chemical potential, $\mu = \left(E_{\mathrm{LUMO}} + E_{\mathrm{HOMO}}\right)/2$, measures the energy change that occurs when an electron is added or removed from a system. Molecules with higher chemical potential values are more likely to accept electrons and have a higher electrophilicity. The chemical hardness, $\eta = \left(E_{\mathrm{LUMO}} - E_{\mathrm{HOMO}}\right)/2$, describes the stability of a molecule and its ability to respond to electron donation or removal. Molecules with lower hardness values are more reactive and have higher electrophilicity. Soft materials have a lesser energy gap than hard materials and more reactive than hard because they can easily provide electrons to an acceptor [55]. Molecular stability and reactivity can also be determined by measuring the chemical softness, $S = 1/2\eta$, which represents the inverse of the resistance of a molecule to undergo electronic transitions. The computed values HOMO and LUMO energies and related quantum parameters are listed in Table 9.

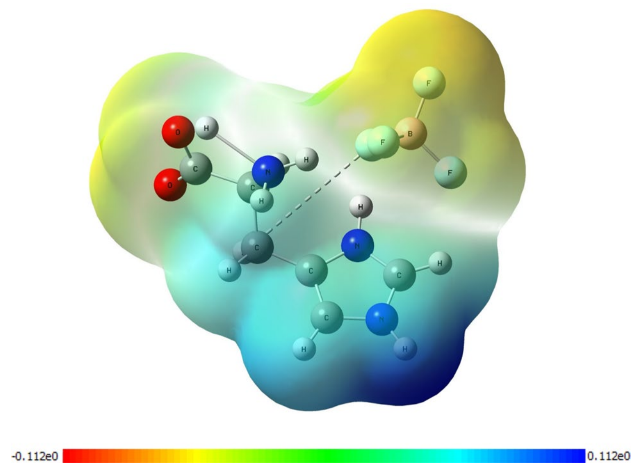


Fig. 16 Molecular electrostatic potential map of LHFB

Due to its calculated wide energy gap and high value of chemical hardness of LHFB, we conclude that this molecule can be classified as a hard molecule which shows good stability and less reactive.

3.11 Molecular electrostatic potential (MEP)

We studied the chemical reactivity of the molecule with the help of the molecular electrostatic potential (MEP) maps which provides the information about the reactive sites of the molecule. This three-dimensional map also gives the visualization of electron acceptor and donor areas. The different colors of MEP map are used to indicate the levels of electrostatic potential at the surface [56]. These include a color gradient from red to blue and potential increases in the order of Red > Orange > Yellow > Green > Blue. The red color shows the areas with the highest density and the most negative electrostatic potential (site for electrophilic attack). Blue color shows the areas with the lowest density and the most positive electrostatic potential (site for nucleophilic attack) and the zero potential region is depicted in green [57]. Figure 16 depicts the calculated MEP map of LHFB along with color code of the map. The color code of this map is found in the range between -0.112 a.u. (Red) and 0.112 a.u. (Blue) for this molecule. In this map, the region with negative electrostatic potential is located over the F and O atoms i.e., potential electrophilic attack site, whereas maximum positive electrostatic potential region is located over the exterior hydrogen atom of the ring, i.e., as potential nucleophilic attack site.

4 Conclusions

L-Histidinium tetrafluoroborate crystals have been grown by a slow evaporation method. Single-crystal XRD confirmed monoclinic lattice with space group $P2_1$. The values of refinement factor (R) and Goodness of Fit (S) of the LHFB structure were found to be 0.0375 and 1.070, respectively. Residual density analysis ensured high-quality structure of LHFB the grown single-crystal. The observed morphology was found to match with the geometrical prediction by BFDH law. The Hirshfeld surface analysis is shown to be a valuable tool to facilitate the identification of different types of intermolecular interactions (H…F, H…H and H…O) and their relative significance to each other. The fingerprint plots reveal. The fingerprint plots showed quantitatively how all the interactions contribute to the crystal packing. The interaction energies represent the relative strength of molecular packing. The enrichment ratios for polar contacts, which are generally hydrogen bonds, were found to be larger than unity. The dielectric constant of grown crystals was found to be 88 at 1 kHz which decreases to 34 at 2 MHz with a low dielectric loss (0.003). The grown crystals depicted very small values of the dielectric constant and dielectric loss, hence they are a promising material for the microelectronic industry.

The piezoelectric property was observed in the amino acid-based LHFB single crystal for the first time in which the piezoelectric charge coefficient was found to be 4 pC/N. The value of Meyer's index number (1.7) has shown the softer nature of LHFB crystal. The volume percentage of the void at 0.002 a.u. in a unit cell of LHFB crystals is 5.72. The pictorial view of the voids and computed volume with surface area revealed the presence of continuous porous channels in the LHFB crystal.

The optimized structural parameter such as bond length, dihedral angle and bond angle are calculated from DFT method. The lowering of HOMO-LOMO energy bandgap support bioactive property of the molecule. Their low value of dielectric loss plus a desirable high piezo-response makes the LHFB single crystals highly suitable for making materials used in optoelectronics, and self-powered devices for energy harvesting and transducer applications.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00339-024-07369-y.

Acknowledgements Dr. Nidhi Tyagi expresses her gratitude to Principal, Shivaji College for encouragement and support for present research work. Dr. Nidhi Sinha would like to thank the Principal, SGTB Khalsa College for encouragement in research work. Dr. Harsh Yadav is thankful to the Vice Chancellor, Netaji Subhas University of Technology for his continuous support to carry out research work and Dr. Pradeep Kumar would like to express his gratitude to the Principal, Hansraj College to providing support for this research work.

Author contributions Conceptualization, investigation, resources, supervision, resources, writing—review and editing were performed by NT, NS and BK. Data collections, methodology, investigations, writing—review and editing were performed by HY and PK. All the authors have read and agreed to the published version of the manuscript.

Funding This research received no external funding.

Data availability Not applicable.

Data and code availability Not applicable.

Declarations

Conflicts of interest The authors declare no conflict of interest.

Ethical approval Not applicable.

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